

Survey Expectations Pipeline

Technical Documentation

*Extrapolating Macroeconomic Survey Expectations to 1920 Using Shiller Macro Features and
WSJ Sentiment*

Aaron Huberman | Duke University
Department of Economics | DCC Cluster

1. Project Overview

Survey-based macroeconomic expectations data — from the Survey of Professional Forecasters (SPF), Michigan Survey, and related sources — only exist from the early 1970s onward for earnings and returns, and from 2003 for dividends. This severely limits long-run analysis of expectations formation and asset pricing.

This pipeline constructs **synthetic survey-consistent expectations** for four series back to 1920 by calibrating ridge regression models on the overlap period where surveys and Shiller macro data coexist, then extrapolating backward using historically available macro observables and WSJ sentiment.

The extrapolated series are intended as inputs to the **Campbell-Shiller variance decomposition**, providing a behaviorally grounded alternative to standard VAR-implied expectations over a century of data.

Four Target Series

Series	Survey Period	Extrapolated Back To	Units
Dividend growth expectations	2003 Q1 – 2022 Q1	1920-01-01	Log growth rate
Earnings growth expectations	1976 Q1 – 2022 Q1	1920-01-01	Log growth rate
1-year S&P 500 return expectations	2000 Q2 – 2024 Q1	1920-01-01	% per year
10-year S&P 500 return expectations	2000 Q2 – 2024 Q1	1920-01-01	% per year (ann.)

2. Data Sources

2.1 Shiller Long-Run Dataset (ie_data.xls)

The backbone of the feature set. Provides monthly S&P 500 price, dividends, earnings, CPI, and 10-year Treasury yield back to 1871. All features are constructed point-in-time at the wave closing date — no look-ahead contamination.

Feature	Derivation	Economic role
dp_ratio	D / P	Dividend yield — direct signal for dividend expectations
ep_ratio	E / P	Earnings yield — primary signal for earnings expectations
cape_inv	1 / CAPE	Cyclically-adjusted earnings yield (available from 1881)
gs10	Rate GS10	Interest rate regime — dominant for return expectations
infl_yoy	CPI 12m %	Nominal vs real inflation environment
real_price_gr	(P/CPI) 12m %	Real market momentum
div_gr_yoy	D 12m %	Realised dividend growth — fundamentals AR(1)
earn_gr_yoy	E 12m % (clipped $\pm 100\%$)	Realised earnings growth — clipped to remove COVID spike
div/earn lag4/8	Shift 3/6 months	Explicit AR(p) lag structure (~ 1 and 2 quarter lags)
earn_gr_chg	diff(earn_gr_yoy, 3)	Earnings revision momentum
trailing_1yr_ret	12m total return	Greenwood-Shleifer extrapolation variable
trailing_3yr_ret	36m total return	Medium-run return extrapolation
dp_dev	dp_ratio minus 10yr mean	Mean-reversion signal for dividend expectations
earn_dev	earn_gr_yoy minus 10yr mean	Mean-reversion signal (clipped $\pm 100\%$)
recession	NBER dummy	1 = NBER recession month, 0 = expansion

2.2 WSJ Sentiment Series (wsj_sentiment_quarterly.csv)

Quarterly sentiment indices constructed from WSJ headlines and article bodies, spanning 1889 Q3 to 2026 Q2. The pipeline uses four columns selected by walk-forward parsimonious search:

Column	Definition	Used for series
cf_tone	Cash-flow tone = (growth words) - (decline words), body-level	Earnings growth, Returns 10yr
earn_salience	Fraction of articles discussing earnings topics	Earnings growth
sent_agg	Composite standardised sentiment index (mean=0, std=0.59)	Returns 1yr

lm_unc	Loughran-McDonald uncertainty word fraction	Returns 10yr
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Sentiment columns are forward-filled from quarterly to monthly frequency to align with the Shiller index. They cover the full survey calibration window for all four series and extend back to 1889, covering the extrapolation period.

2.3 Survey Data Files

File	Key columns used
Dividend_growth_expectations.xlsx	Expected one-year log dividend growth, Realized next year log dividend growth
Earnings_growth_expectations.xlsx	Col 2 (expected, current earnings denominator), Col 3 (realized)
return_expectations_quarterly.xlsx	sp_1yr_median, sp_10yr_median (% per year)

3. Model Architecture

3.1 Common Structure

All four series use the same two-step sklearn Pipeline: **StandardScaler** (zero mean, unit variance) followed by **RidgeCV** (L2-regularised linear regression with 5-fold cross-validation alpha selection over 50 log-spaced values from 10^{-3} to 10^4). Ridge is preferred over Random Forest because RF boundary averaging produces implausibly muted extrapolations in historical regimes not seen during calibration.

Alpha is re-selected at each walk-forward step from the same grid, so the regularisation strength adapts as the calibration window grows. The final model is re-fitted on all available survey waves for use in extrapolation.

3.2 Per-Series Feature Sets and Results

Feature sets were selected by parsimonious walk-forward search: starting from theoretically motivated macro features, adding sentiment and regime variables one at a time and retaining additions that improved both walk-forward R^2 and subsample stability. The stability metric is the Pearson correlation between Ridge coefficients fitted on the first vs second half of the calibration window.

Dividend Growth Expectations

Walk-forward R^2	Subsample stability	Eval steps	Features	Ridge alpha	Special feature
+0.606	0.795	52	18 macro + survey lag	1e5	Survey lag

Feature set:

- dp_ratio, ep_ratio, cape_inv, gs10, infl_yoy, real_price_gr
- div_gr_yoy, earn_gr_yoy
- div_gr_lag4, div_gr_lag8, earn_gr_lag4, earn_gr_lag8
- earn_gr_chg, div_gr_chg
- trailing_1yr_ret, trailing_3yr_ret
- dp_dev, earn_dev
- + survey_lag1 (prior quarter survey value)

Rationale: Dividend survey expectations move *opposite* to realised dividend growth, consistent with boards targeting long-run payout ratios. The dp_dev feature captures mean-reversion directly. Sentiment features were tested but hurt stability (r dropped to 0.07–0.33); macro-only wins. The survey lag (prior quarter survey value) adds +0.468 R^2 with stability improving to 0.795. During pre-survey extrapolation the lag is bootstrapped: each prediction is fed as the lag for the next month.

Extrapolation note: 80% of pre-survey months are OOD (calibration window only 2003–2022). Early extrapolation carries highest uncertainty.

Earnings Growth Expectations

Walk-forward R^2	Subsample stability	Eval steps	Features	Ridge alpha	Special feature
+0.707	0.941	145	6	~0.19	Recession dummy

Feature set:

- ep_ratio — earnings yield (primary signal)
- cape_inv — cyclically-adjusted earnings yield
- real_price_gr — real market momentum
- cf_tone — WSJ cash-flow tone (growth minus decline words)
- earn_salience — fraction of WSJ articles discussing earnings
- recession — NBER recession dummy

Rationale: The most stable and accurate series. ep_ratio dominates with a large negative coefficient — when valuations are high, earnings growth expectations are lower. The parsimonious 3-feature macro set already achieved $R^2=0.685$; cf_tone and earn_salience add +0.012 with only minor stability cost. The recession dummy adds +0.010 with stability actually improving to 0.941, making it the most credible series for extrapolation. Earnings outliers (COVID snap-back of 790%) are clipped at $\pm 100\%$ before feature construction.

Extrapolation note: Only 18% of extrapolation window is OOD — longest calibration window (1976–2022). Most reliable series for pre-survey period.

1-Year S&P; 500 Return Expectations

Walk-forward R^2	Subsample stability	Eval steps	Features	Ridge alpha	Special feature
+0.108	0.905	66	4	~72	None (recession hurt)

Feature set:

- gs10 — 10-year Treasury rate (dominant signal)
- dp_ratio — dividend yield
- infl_yoy — CPI year-on-year inflation
- sent_agg — composite WSJ sentiment index

Rationale: Short-run return expectations are difficult to model — even the best literature models struggle. The recession dummy hurt R^2 (-0.034) so was excluded. sent_agg flips R^2 from -0.070 to +0.108, consistent with short-run expectations being more sentiment-driven than long-run. The model beats AR(1) on RMSE by 0.185 despite modest R^2 . High alpha (72) indicates aggressive regularisation — Ridge is close to predicting the mean, with features providing mild adjustments.

Extrapolation note: 72% of extrapolation window OOD (calibration only 2000–2024). Pre-survey predictions outside [3.6, 7.4] should be treated with caution.

10-Year S&P; 500 Return Expectations

Walk-forward R ²	Subsample stability	Eval steps	Features	Ridge alpha	Special feature
+0.176	0.693	66	6	~14	Recession dummy

Feature set:

- gs10 — 10-year Treasury rate (dominant signal)
- ep_ratio — earnings yield
- dp_ratio — dividend yield
- lm_unc — LM uncertainty word fraction
- cf_tone — WSJ cash-flow tone
- recession — NBER recession dummy

Rationale: Long-run return expectations are dominated by gs10 (positive coefficient) and valuation ratios ep_ratio and dp_ratio (negative coefficients), consistent with the Gordon Growth Model intuition. lm_unc and cf_tone add marginal improvement. The recession dummy adds +0.029 R² with stable coefficients. Subsample stability (0.693) is the weakest of the four series, reflecting genuine structural change in the relationship between macro conditions and long-run return expectations across different interest rate regimes.

Extrapolation note: 72% OOD. In-distribution predictions range [4.9, 9.2]% — OOD months may produce implausible negatives.

4. Evaluation Methodology

4.1 Walk-Forward Expanding Window

All models are evaluated via **walk-forward expanding-window cross-validation**. At each step t , the model is trained on waves 0 through $t-1$ and predicts wave t . No future information ever enters training. This scheme avoids regime-shift contamination and produces a realistic simulation of out-of-sample performance.

Parameter	Dividend growth	Earnings growth	Returns 1yr	Returns 10yr
Min training waves	25	40	30	30
Walk-forward eval steps	52	145	66	66
Alpha selection CV	5-fold	5-fold	5-fold	5-fold

4.2 Metrics

Metric	Definition	Interpretation
WF R^2	$r^2(\text{actuals}, \text{walk-forward predictions})$	0 means model beats predicting the mean. Primary metric.
WF RMSE	$\sqrt{\text{mean}((\text{actual} - \text{predicted})^2)}$	Absolute prediction error in original units.
AR(1) RMSE	$\sqrt{\text{mean}((\text{actual} - \text{prior survey wave})^2)}$	Naive persistence baseline. Model should beat this.
vs AR(1)	WF RMSE - AR(1) RMSE	Negative = model beats naive persistence. Key benchmark.
Stab r	Pearson corr of Ridge coefficients, $\text{cal} \rightarrow \text{out}$	>0.9 stable, <0.7 unstable. Predicts extrapolation credibility.

4.3 Out-of-Distribution Flagging

Pre-survey extrapolations are flagged using **per-series Mahalanobis distance** in each series' own feature space. The threshold is the 97.5th percentile of Mahalanobis distances in the calibration data. Months exceeding this threshold are marked OOD in the output CSV and plotted as red dots.

OOD flags do not exclude predictions — they are warning labels. For the Campbell-Shiller decomposition, consider either using only in-distribution months or applying wider uncertainty bands to OOD periods.

5. Pipeline Execution

5.1 File Structure

```
survey_expectations_pipeline/ ■■■■ macro_expectations_pipeline.ipynb ■■■■ README.md ■■■■
data/ ■ ■■■■ ie_data.xls ■ ■■■■ Dividend_growth_expectations.xlsx ■ ■■■■
Earnings_growth_expectations.xlsx ■ ■■■■ return_expectations_quarterly.xlsx ■ ■■■■
wsj_sentiment_quarterly.csv ■■■■ output/ ← created on first run
```

5.2 Cell Execution Order

Cell	Section	What it does
1	Imports	pandas, numpy, sklearn, matplotlib
2	Config	Paths, MIN_TRAIN_WAVES, SERIES_FEATURES, NBER_RECESSIONS
3-4	Load Shiller	Parse ie_data.xls, construct all 18 macro features + recession dummy
5-6	Load Sentiment	Read wsj_sentiment_quarterly.csv, forward-fill to monthly Shiller index
7-8	Load Surveys	Read all four survey Excel files, construct wave_date columns
9-10	Align	Match each survey wave to last available Shiller month
11-12	Model	Define make_model() — StandardScaler + RidgeCV(cv=5)
13-14	Walk-forward	Define walk_forward() with survey lag support for dividends
15-16	Run	Execute walk-forward for all series, fit final models
17	Summary table	Print WF R ² , RMSE, AR(1) comparison
18-19	Diagnostic plots	3-panel time series + scatter + rolling RMSE per series
20-21	Feature importance	Standardised Ridge coefficients bar chart
22-23	Save models	Pickle final pipelines to output/{series}/model_final.pkl
24-25	Extrapolation	Build 1920→today feature matrix, per-series OOD, bootstrap lag
26	Full-range plot	Combined 4-panel time series from 1920 to today
27-28	Stability check	Subsample coefficient correlation early vs late
29-32	Summary	Final results and output file paths

5.3 Configuration

All key parameters are in cell 2. To run on the DCC cluster:

```
BASE_DIR = Path("/hpc/home/ah620") DATA_DIR = BASE_DIR / "data" OUTPUT_DIR = BASE_DIR /
"output"
```

To change the extrapolation window:

```
EXTRAP_START = "1920-01-01" # EXTRAP_END defaults to shiller_df.index[-1] (latest available data)
```

To add a new series to USE_SURVEY_LAG (bootstrapped lag during extrapolation):

```
USE_SURVEY_LAG = {"dividend_growth"} # add series names here
```

6. Outputs

6.1 Extrapolated Expectations CSVs

One CSV per series at `output/{series}/extrapolated_expectations.csv`

Column	Type	Description
date	datetime	Month start date, monthly frequency, 1920-01-01 → latest Shiller
predicted	float	Model-predicted expectation in original survey units
ood	bool	True if Mahalanobis distance exceeds 97.5th pct of calibration

6.2 Plots

File	Description
full_range_expectations.png	4-panel: model prediction + survey actual, 1920–2026
expectations_vs_realized.png	4-panel: adds realized outcomes in red
diagnostics_{series}.png	3-panel: walk-forward predictions, scatter, rolling RMSE
feature_importance.png	Standardised Ridge coefficients for all series

6.3 Saved Models

Each series has a pickled sklearn Pipeline at `output/{series}/model_final.pkl` containing:

- **model** — fitted Pipeline (StandardScaler + RidgeCV)
- **features** — list of feature names the model expects
- **wf_r2**, **wf_rmse**, **ar1_rmse**, **dir_acc** — walk-forward metrics
- **n_waves** — number of survey waves used for final fit
- **saved_at** — ISO timestamp

6.4 Key Caveats for Campbell-Shiller Decomposition

The extrapolated series should be treated as **noisy proxies** for true historical expectations, not as ground truth. Specific cautions:

- **Dividend growth (pre-2003)**: 80% OOD, bootstrapped lag compounds uncertainty. Treat with widest error bands.
- **Return series (pre-2000)**: 72% OOD. Clip or exclude OOD months with implausible negative values for the decomposition.
- **Earnings growth (pre-1976)**: Most credible at 18% OOD. Preferred series for long-run decomposition claims.

- **Structural breaks:** Subsample stability below 0.7 (returns 10yr) indicates the calibrated relationship may not be stable across interest rate regimes.
- **Regime assumption:** All models assume the mapping from macro conditions to survey expectations is stable across the full extrapolation window. This is testable but not guaranteed.